

SEMINAR ON DKA MANAGEMENT

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Outline

- Introduction
- Pathophysiology Of DKA
- Clinical feature Of DKA
- Diagnosis of DKA
- Investigation of DKA
- Differential Diagnosis
- General management and approach Of DKA
- Fluid and electrolyte management
- Insulin therapy
- Monitoring
- Complication during management and prevention

INTRODUCTION

Diabetes Mellitus (DM)

- Diabetes Mellitus is not a single disease entity.
- It is a group of common chronic metabolic disorders sharing a common underlying feature of hyperglycemia.
- It is **progressive disease** effecting numerous aspects of metabolism, beyond those related to glucose homeostasis
- Hyperglycemia in DM results from defects in insulin secretion, insulin action, or, both

Etiologic Classifications of DM

- Reflects the major **etiologies** and **pathogenic** pathways that are thought to lead to diabetes
- The **ADA** has proposed a diabetes classification system in 4 categories:
 - type 1 diabetes,
 - type 2 diabetes,
 - other specific types, and
 - Gestational DM

Type 2 Diabetes Mellitus

- Due to insulin resistance with relative insulin deficiency
- Associated with:
 - Obesity
 - Family history
 - Sedentary lifestyle
- The incidence of T2DM among youth is increasing in many countries

Other Specific Types

- A. Genetic Defects Of β -Cell Function and Insulin Action
- B. Other Genetic and Autoimmune Syndromes Associated With Diabetes
- C. Drug Or Chemical Induced and Infections
- D. Diseases of E.Pancreas and Associated Endocrinopathies

Type 1 Diabetes Mellitus

- Most common form in children
- Caused by autoimmune destruction of pancreatic β -cells
- Results in absolute insulin deficiency

Epidemiology

- T1DM is the most common in childhood and adolescence.
- It accounts for approximately 10% of all cases of diabetes in all ages.
- F=M
- Peaks of presentation occur in two age- groups: at 5- 7 (the time of increased exposure to infectious agents) and, at puberty, due to increasing in gonadal steroids and GH secretion (which antagonizes Insulin),

Pathogenesis of T1DM

GENETIC PREDISPOSITION

HLA-linked genes and
other genetic loci



Immune response against
normal beta cells

AND/OR

Immune response against
altered beta cells

ENVIRONMENTAL INSULT

Viral infection:
Molecular mimicry

AND/OR

Damage to beta cells



AUTOIMMUNE ATTACK

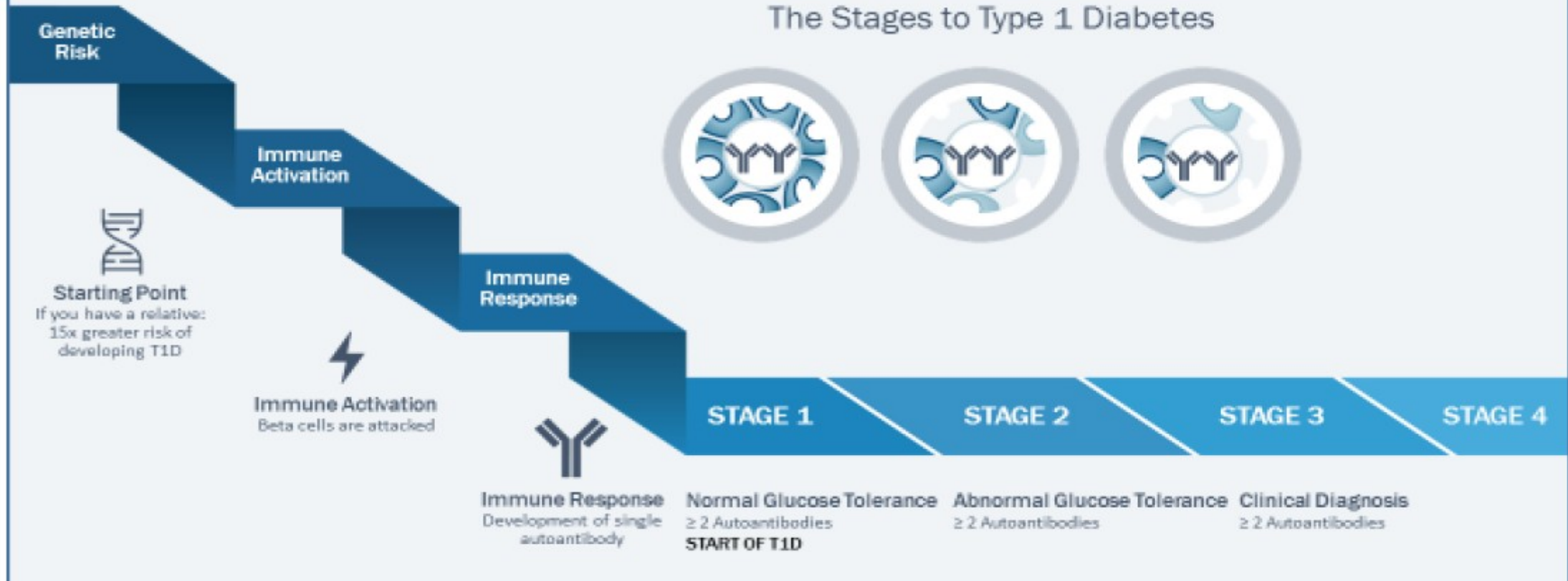
Beta-cell destruction



TYPE 1 DIABETES

Natural History of Type 1 Diabetes Mellitus

Type 1 Diabetes Disease Progression



Criteria for the Diagnosis of Diabetes Mellitus

Table 629.2	Diagnostic Criteria for Dysglycemia and Diabetes Mellitus
DYSGLYCEMIA	DIABETES MELLITUS
IMPAIRED FASTING GLUCOSE Fasting (at least 8 hr) plasma glucose 100–125 mg/dL (5.6–7.0 mmol/L)	Fasting (at least 8 hr) plasma glucose $\geq$ 126 mg/dL (7.0 mmol/L) Or
IMPAIRED GLUCOSE TOLERANCE 2-hr plasma glucose during OGTT $\geq$ 140 mg/dL (7.8 mmol/L), but <200 mg/dL (11.1 mmol/L)	2-hr plasma glucose during OGTT $\geq$ 200 mg/dL (11.1 mmol/L) Or
PREDIABETES Hemoglobin A _{1c} 5.7–6.4% (39–47 mmol/mol)	Hemoglobin A _{1c} $\geq$ 6.5% (48 mmol/mol) Or Symptoms* of diabetes mellitus plus random or casual plasma glucose $\geq$ 200 mg/dL (11.1 mmol/L) [†]

*Symptoms include polyuria, polydipsia, and unexplained weight loss with glucosuria and ketonuria.

[†]Results should be confirmed by repeat testing if in absence of unequivocal hyperglycemia.

OGTT, Oral glucose tolerance test.

Current ADA diagnostic criteria for diabetes mellitus

Diagnostic test	Normal	Pre-diabetic	diabetic	Remark
Fasting blood glucose (FBS) □	75 to < 100 mg/dl	100-125mg/dl	>126 mg/dl	At least 2 tests needed
Hemoglobin A1C (HbA1C)	4.0 – 5.6%	5.7-6.4%	> 6.5%	At least 2 tests needed
02-hour plasma glucose tolerance test (OGTT)	< 140 mg/dl	140-199mg/dl	>200mg/dl	At least 2 tests needed
Random blood glucose (RBS)	65 to < 140 mg/dl	140-199mg/dl	>200mg/dl	DM diagnosed Only if RBS >200mg/dl is accompanied by Classic Symptoms of DM

Complications of DM

Acute complications:

- ▶ Diabetic Ketoacidosis
- ▶ Hypoglycemia
- ▶ HHS

Chronic complications:

- ▶ Microvascular
- ▶ Macrovascular

Diabetic Ketoacidosis (DKA)

Diabetic ketoacidosis (DKA) is an acute, life-threatening complication of diabetes mellitus.

- Formerly considered a hallmark of type 1 DM.

 - also occurs in individuals with type 2 DM

 - ✓ Ketosis-prone diabetes mellitus

- Characterized by:-

 - ✓ hyperglycemia

 - ✓ ketonemia

 - ✓ metabolic acidosis

Epidemiology

- DKA is the most common acute complication of T1DM in children.
- 20–30% of newly diagnosed type 1 diabetes cases in children present with DKA.
- Major cause of death: cerebral edema (accounts for 60–90% of pediatric DKA deaths).
- Higher mortality in low-resource settings due to delayed diagnosis, limited ICU access, and infections.

RISK FACTORS

Newly diagnosed patients include

Factors that increase DKA at the initial presentation of T1DM include :

- Young age (<5 years of age and especially <2 years)
- Delayed diagnosis of diabetes
- Low socioeconomic status, lack of health insurance
- Children living in countries with low prevalence of T1DM

Cont...

Known diabetic patient includes

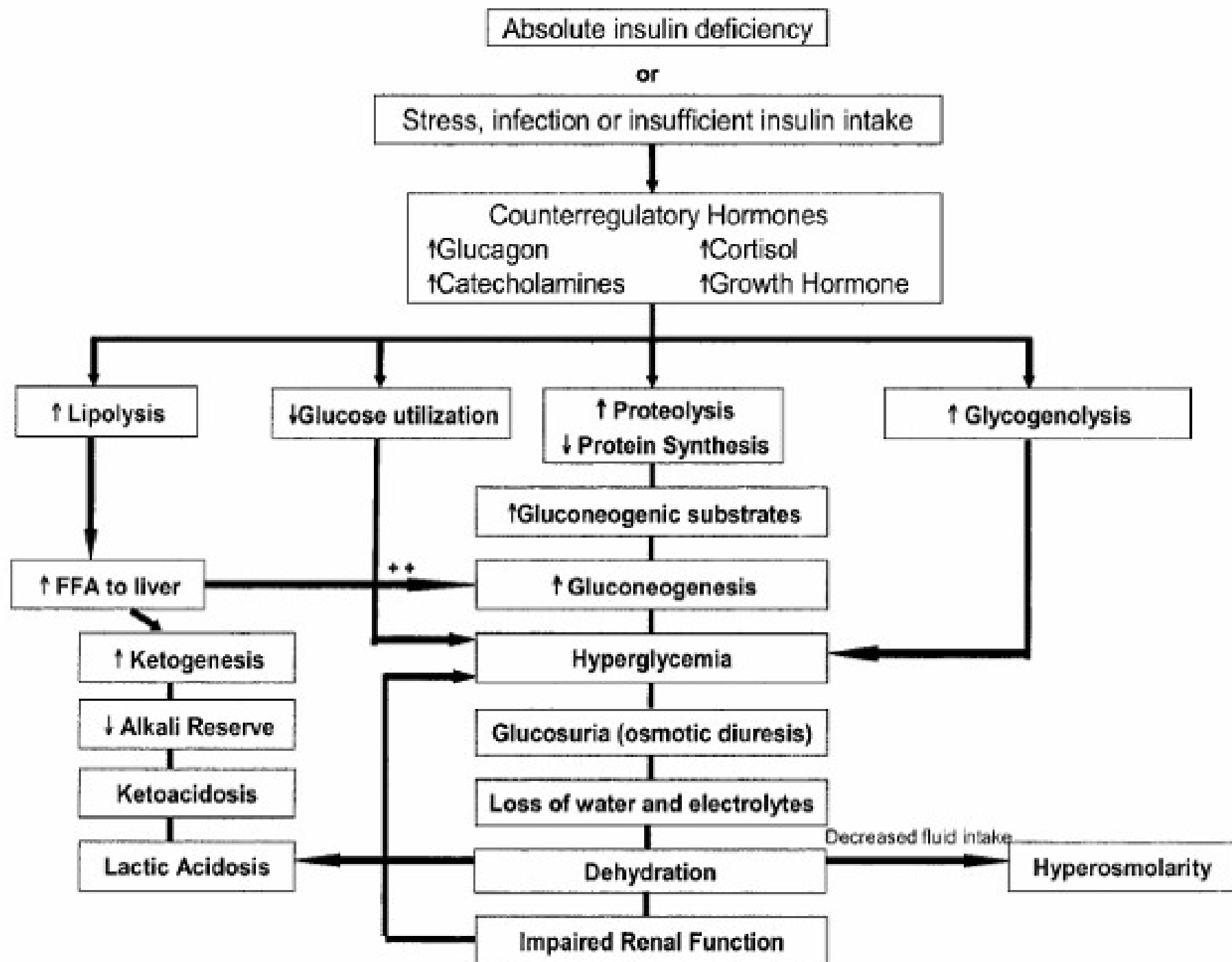
- ✓ **Omission of insulin** for various reasons
- ✓ Children with poor metabolic control or previous episodes of DKA
- ✓ **Gastroenteritis** with persistent vomiting and inability to maintain hydration
- ✓ Children with **psychiatric disorders**, including those with eating disorders
- ✓ Difficult or unstable family circumstances (eg, parental abuse)
- ✓ Children with limited access to medical services

Pathophysiology of DKA

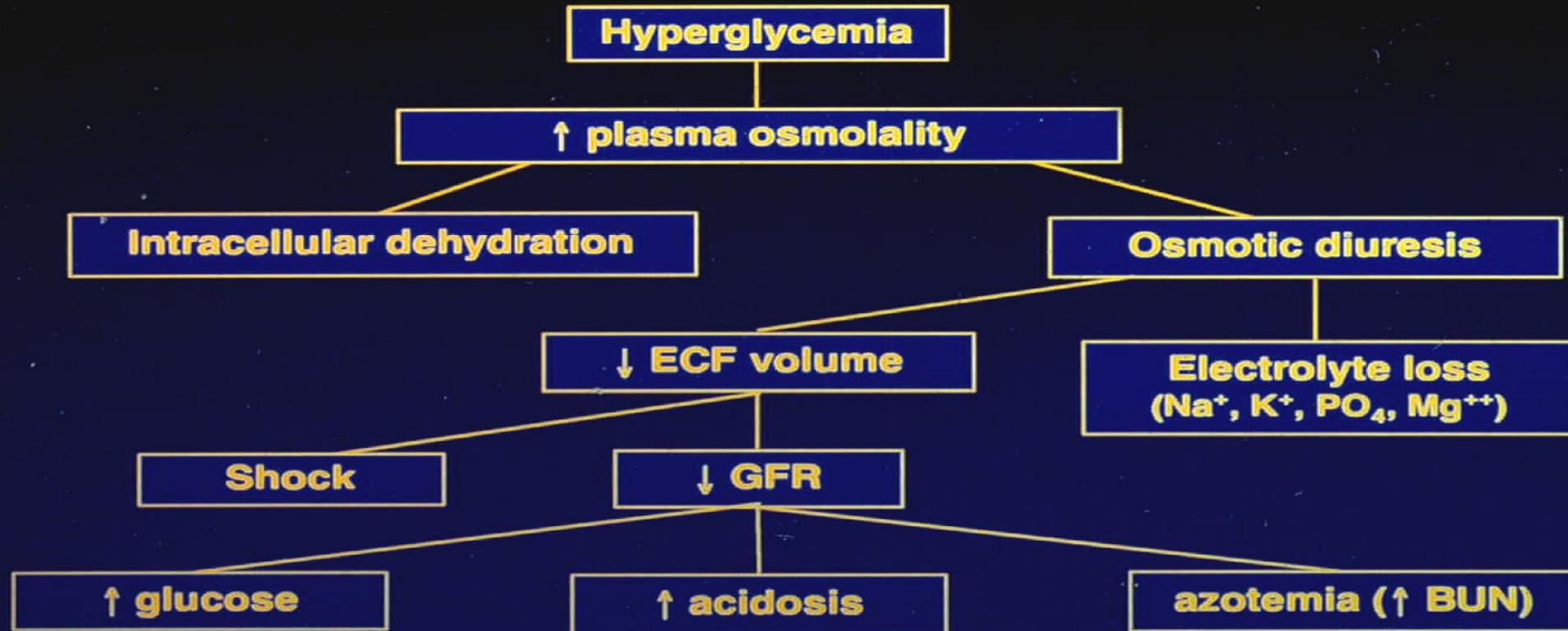
- results from relative or absolute insulin deficiency combined with counterregulatory hormone excess.
- Both insulin deficiency and glucagon excess, in particular, are necessary for DKA to develop.

Cont...

- The decreased ratio of insulin to glucagon promotes;
 - ✓ gluconeogenesis,
 - ✓ glycogenolysis, and
 - ✓ ketone body formation in the liver, as well as
 - ✓ substrate delivery from fat and muscle (free fatty acids, amino acids) to the liver.



Diabetic Ketoacidosis



Euglycemic DKA

Euglycemic ketoacidosis occurs with glucose level < 200mg/dl. Situations in which euDKA occurs include:-

1. Patients presenting shortly after receiving insulin
2. Type1diabetics who are young and vomiting
3. Patients with impaired gluconeogenesis (alcohol abuse or liver failure)
4. Low caloric intake/starvation
5. Pregnancy

Approch to DKA

- Approach to the patients starts by
 - ✓ History taking and physical physical examination guide us to diagnosis DKA and HHS.
 - ✓ Confirm it with lab findings.

Clinical Features of DKA

- Signs and symptoms of DKA are the result of acidosis, hyperglycemia, volume depletion, and electrolyte losses.

TABLE 397-7 Manifestations of Diabetic Ketoacidosis

Symptoms	Physical Findings
Nausea/vomiting	Tachycardia
Thirst/polyuria	Dehydration/hypotension
Abdominal pain	Tachypnea/Kussmaul respirations/ respiratory distress
Shortness of breath	Abdominal tenderness (may resemble acute pancreatitis or surgical abdomen)
Precipitating events	Lethargy/obtundation/cerebral edema/possibly coma
Inadequate insulin administration	
Infection (pneumonia/UTI/ gastroenteritis/sepsis)	
Infarction (cerebral, coronary, mesenteric, peripheral)	
Drugs (cocaine)	
Pregnancy	

Abbreviation: UTI, urinary tract infection.

Cont..

Infants tend to present with

- decreased energy and activity,
- Vomiting
- Irritability or Lethargy
- weight loss, and physical signs of dehydration; a severe Candida diaper rash is common.

The biochemical criteria for the diagnosis of DKA

- The diagnosis of DKA is based on the triad of **hyperglycemia, ketosis and metabolic acidosis**

➤ All three biochemical criteria are required to diagnose DKA

1. Hyperglycemia (blood glucose >11 mmol/L [200 mg/dL])
2. Venous pH <7.3 or serum bicarbonate <18 mmol/L
3. Ketonemia (BOHB ≥ 3 mmol/L) or moderate or large ketonuria.

Typical laboratory characteristics of DKA and HHS*

	DKA			HHS
	Mild	Moderate	Severe	
Plasma glucose (mg/dL)	>250	>250	>250	>600
Plasma glucose (mmol/L)	>13.9	>13.9	>13.9	>33.3
Arterial pH	7.25 to 7.30	7.00 to 7.24	<7.00	>7.30
Serum bicarbonate (mEq/L)	15 to 18	10 to <15	<10	>18
Urine ketones [¶]	Positive	Positive	Positive	Small
Serum ketones - Nitroprusside reaction	Positive	Positive	Positive	≤ Small
Serum ketones - Enzymatic assay of beta hydroxybutyrate (normal range <0.6 mmol/L) ^Δ	3 to 4 mmol/L	4 to 8 mmol/L	>8 mmol/L	<0.6 mmol/L
Effective serum osmolality (mOsm/kg) [◇]	Variable	Variable	Variable	>320
Anion gap [§]	>10	>12	>12	Variable
Alteration in sensoria or mental obtundation	Alert	Alert/drowsy	Stupor/coma	Stupor/coma

Cont..

- Differences in treatment strategy between HHS and DKA include
 - ✓ volume of fluid administered
 - ✓ the timing of administration of insulin
 - ✓ and monitoring of the corrected sodium decline

Severity of DKA (ISPAD 2022)

- Is categorized by the degree of acidosis

Mild:

- ✓ venous pH <7.3 or
- ✓ Serum bicarbonate <18 mmol/L₂₄

Moderate:

- ✓ pH <7.2 or
- ✓ Serum bicarbonate <10 mmol/L

Severe:

- ✓ pH <7.1 or
- ✓ Serum bicarbonate <5 mmol/L

Classification of Diabetic Ketoacidosis

	NORMAL	MILD	MODERATE	SEVERE*
CO ₂ (mEq/L, venous) †	20-28	16-20	10-15	<10
pH (venous) †	7.35-7.45	7.25-7.35	7.15-7.25	<7.15
Clinical	No change	Oriented, alert but fatigued	Kussmaul respirations; oriented but sleepy; arousable	Kussmaul or depressed respirations; sleepy to depressed sensorium to coma

* Severe hypernatremia (corrected Na >150 mEq/L) would also be classified as severe diabetic ketoacidosis.

† CO₂ and pH measurement are method dependent; normal ranges may vary.

Investigations

- ✓ Blood glucose
- ✓ Blood (BOHB)
- ✓ Urine ketone
- ✓ Electrolytes (including serum bicarbonate)
- ✓ Blood urea nitrogen, creatinine
- ✓ Serum osmolality
- ✓ Venous pH, pCO₂
- ✓ Complete blood count (Hgb, Hct)
- ✓ Culture (blood, urine, throat)

Differential Diagnosis

- Alcoholic ketoacidosis
- Uremia
- Lactic acidosis
- Sepsis
- Hypoglycemia,
- Hyperosmolar hyperglycemic state
- Starvation ketoacidosis,
- Renal failure
- Ingestions such as salicylate, methanol, ethylene glycol, iron, or isoniazid.

Management of Diabetes Mellitus:

Can be approached in 3 Phases depending on the initial presentation; all of which have separate goals but in practice they merge in to a continuum.

- ☐ Phase I :-DKA Management
- ☐ Phase II:-Post acidotic or transition period for metabolic control.
- ☐ Phase III:-The continuation phase with guidance of the Diabetic child and his or her family.

DKA Management

Where should the child with DKA be managed?

- Experienced nursing and medical staff trained in pediatric DKA management
- Care policies and procedures based on clinical practice guidelines
- Access to a laboratory appropriate for a standard of care
- Home vs ER vs General ward vs PICU
 - Severity of DKA
 - Established vs new-onset DM
 - Proficiency of caretakers
 - Vomiting and feeding tolerance
 - Institutional guidelines

Principles of Treatment

- ABC of life
- Careful replacement of fluid deficits.
- Correction of acidosis & hyperglycemia
- Correction of electrolytes imbalance.
- Treatment of complications
- Treatment of precipitating factor
- Diabetic education
- Monitoring and followup

The goals of therapy

- Restore perfusion and correct dehydration
- Stop Ketogenesis
- Restore blood glucose to near normal
- Correct electrolyte losses
- Avoid complications of the Rx
- Identify and treat any precipitating event

Fluid and electrolyte replacement

The objectives of fluid and electrolyte replacement therapy are to:

- ✓ Restore circulating volume
- ✓ Replace sodium and water deficits
- ✓ Improve glomerular filtration and
- ✓ Enhance clearance of glucose and ketones from the blood

Cont...

- ✓ The general approach and principles of management are the same for all children with DKA, regardless of the severity.
- ✓ Patients with DKA have a deficit in (ECF) volume that is typically about 7% of body weight
- ✓ Shock with hemodynamic compromise is rare in pediatric DKA
- ✓ Clinical estimates of the degree of dehydration are inaccurate .
- ✓ Instead, patients should be rehydrated based upon a presumed fluid deficit and clinical response

Cont..

Recommendations based on PECARN FLUID Trial:

- ✓ Assumed fluid deficits between 5%-10% of body weight
- ✓ Replacement of deficits over 24 to 48 hours
- ✓ Provision of maintenance fluids
- ✓ Use of fluids with a sodium content between 0.45% and 0.9% NaCl

Cont...

Resuscitation fluids: initial fluid resuscitation

- Initial volume expansion of **10 to 20 mL/kg** should be administered as an IV bolus, using **isotonic** saline or Ringer's lactate infused over **20 to 30 minutes**.
- If circulating volume is **still compromised** after the initial bolus is complete, additional IV bolus infusions of **10 to 20 mL/kg** can be given.
- *10-20 ml/kg NS for iv volume depletion but **not shock***
- **20ml/kg for shock**

Cont...

Deficit replacement

- Over 24-48
- NS
- Change fluid to D5 containing when the plasma glucose falls to 250-300 mg/dl.

Subsequent fluid administration —

- Once the child is hemodynamically stable, additional IV fluids should be administered, using IV fluids with 0.45 to 0.9% NaCl.
- The rate of fluid administration is calculated to replace the remaining fluid deficit over 24 to 48 hours.

Diabetic Ketoacidosis Treatment Protocol

TIME	THERAPY	COMMENTS
1st hr	10-20 mL/kg IV bolus 0.9% NaCl or LR Insulin drip at 0.05 to 0.10 units/kg/hr	Quick volume expansion; may be repeated. NPO. Monitor I/O, neurologic status. Use flow sheet. Have mannitol at bedside; 1 g/kg IV push for cerebral edema
2nd hr until DKA resolution	0.45% NaCl; plus continue insulin drip 20 mEq/L KPhos and 20 mEq/L KAc 5% glucose if blood sugar <250 mg/dL (14 mmol/L)	$\text{IV rate} = \frac{85 \text{ mL/kg} + \text{maintenance} - \text{bolus}}{23 \text{ hr}}$
		If K <3 mEq/L, give 0.5-1.0 mEq/kg as oral K solution <i>or</i> increase IV K to 80 mEq/L
Variable	Oral intake with subcutaneous insulin	No emesis; CO ₂ ≥16 mEq/L; normal electrolytes
Note that the initial IV bolus is considered part of the total fluid allowed in the 1st 24 hr and is subtracted before calculating the IV rate.		
Maintenance (24 hr) = 100 mL/kg (for the first 10 kg) + 50 mL/kg (for the second 10 kg) + 25 mL/kg (for all remaining kg)		
Sample calculation for a 30-kg child:		
1st hr = 300 mL IV bolus 0.9% NaCl or LR		
$\text{2nd and subsequent hr} = \frac{(85 \text{ mL} \times 30) + 1750 \text{ mL} - 300 \text{ mL}}{23 \text{ hr}} = \frac{175 \text{ mL}}{\text{hr}}$ (0.45% NaCl with 20 mEq/L KPhos and 20 mEq/LKAc)		

DKA, diabetic ketoacidosis; I/O, input and output (urine, emesis); K, potassium; KAc, potassium acetate; KPhos, potassium phosphate; LR, lactated Ringer solution; NaCl, sodium chloride; NPO, nothing by mouth.

Potassium replacement

- Children with DKA have total body potassium deficits on the order of 3 to 6 mmol/kg
- The major loss of potassium is from the intracellular pool transcellular shifts caused by
 - ✓ hypertonicity
 - ✓ urinary potassium excretion
 - ✓ vomiting

Cont...

- Potassium replacement is required regardless of the serum potassium concentration , except if renal failure is present.
- If patient is hypokalemic, start potassium replacement *at the time of* initial volume expansion and before starting insulin therapy .
- ✓ using a separate iv infusion with a potassium concentration of 40 mmol/L.

Cont...

- Otherwise, start replacing potassium *after* initial volume expansion and concurrent with starting insulin therapy
- If patient is hyperkalemic, it is recommended that Potassium is only added to IV fluids after the patient;
 - ✓ Has passed urine
 - ✓ gives a history of having recently passed urine
 - ✓ after the Potassium has fallen to within the upper limit of the normal range

Cont...

- **Potassium assessment**
- Total body potassium is **always depleted** in DKA
- If serum K^+ result is delayed → **perform baseline ECG**
 - Tall peaked T waves → hyperkalemia
 - Flat T waves / U waves → hypokalemia

Phosphate

- Phosphate depletion occurs in DKA due to urinary loss from osmotic diuresis,
- reduced renal tubular reabsorption of phosphate and a shift of intracellular phosphate to the extracellular compartment as a result of metabolic acidosis

Cont...

- During treatment, 50%–60% of children develop hypophosphatemia.
- Severe hypophosphatemia (<1 mg/dl [0.32 mmol/L]) with or without associated symptoms should be treated promptly

Sodium

- Promoting a rise in serum sodium concentrations during DKA treatment is no longer considered necessary
- Corrected Na = measured Na + $1.6([\text{plasma glucose} - 5.6]/5.6)$ mmol/L or measured Na + $1.6([\text{plasma glucose} - 100]/100)$ mg/dL
- As the plasma glucose concentration falls, measured and corrected sodium should rise 1.6 mmol/L for each 100 mg/dL.

Bicarbonate

- Bicarbonate administration is not recommended except for treatment of life-threatening
- hyperkalemia or for severe acidosis (venous pH <6.9) with evidence of compromised cardiac contractility

Insulin

- Start insulin infusion 1 hr after initiation of IV fluid treatment
- Dose: 0.05-0.1 U/kg/hour of regular insulin
lower dosage (0.05 U/kg/hour) can be considered for children with pH>7.15
(e.g. dilute 50 units regular [soluble] insulin in 50 mL normal saline, 1 unit = 1 mL)
- Route of administration: Intravenous (IV)
- Infusion tubing should be flushed with 20 ml insulin solution before administration.

- The dose of insulin should usually remain at 0.05-0.1 unit/kg/hour at least until resolution of DKA
 - pH >7.30,
 - serum bicarbonate >18 mmol/L,
 - BOHB <1 mmol/L,
 - or closure of the anion gap
- If the insulin effect is adequate, serum BOHB should decrease by approximately 0.5 mmol/L per hour

If iv infusion is not possible

- hourly or 2-hourly SC rapid acting insulin analog (insulin lispro or insulin aspart) is used
- Dose SC: 0.15 units/kg every 2 hours
- The dose can be reduced to 0.1 unit/kg every 2 hours if BG continues to decrease by >5 mmol/L (90mg/dL) even after adding dextrose

Monitoring

- Hourly monitoring
 - Vital sign
 - Neurologic assessment
 - Insulin administration
 - Fluid balance
 - Capillary blood glucose
- Every 2-4 hr
 - Serum electrolytes, calcium, magnesium, BUN, phosphate, and blood gas
 - Blood BOHB concentrations/Urine ketone

DKA/HHS STATE FLOW SHEET

Name_____	Birthdate_____	Admission date/time_____	Weight_____	Height_____	Duration of diabetes_____
Time					
Hrs:min since administered					
Signs					
Temp					
Pulse					
Blood pressure					
Respiratory rate					
Glasgow coma scale					
Pupil size					
Pupil reaction					
Cap refill time					
Laboratory					
Glucose					
pH					
Sodium					
Potassium					
Chloride					
CO ₂					
HCO ₃					
BUN/creatinine					
Ketones serum					
Ketones urine					
β-Hydroxybutyrate					
Osmolality measured					
Osm calculated					
Calcium/phosphorus/magnesium					
Fluids and Insulin					
Type					
Potassium mEq/L					
Rate					
Intake					
Urine output					
Insulin units/hour					

Resolution of DKA

- Serum anion gap reduced to normal (12 ± 2 mEq/L) or serum BOHB ≤ 1 mmol/L (10.4 mg/dL)
- Venous pH > 7.3 or serum bicarbonate > 15 mEq/L
- Blood glucose < 200 mg/dL (11.1 mmol/L)
- Patient is tolerating oral intake

Transition

- Oral fluids should be introduced only when substantial clinical improvement has occurred (mild acidosis/ketosis may still be present)
- When oral fluid is tolerated, IV fluid should be reduced.
- In patients with insulin infusion to prevent rebound hyperglycemia the first SC injection should be given 1-2 hours (with Regular insulin) before stopping the insulin infusion.
- In our case you can adjust the time for the initiation of standing dose, until that give $\frac{1}{2}$ of the six hourly regular insulin dose.

Complications during DKA management

- Hypoglycemia
- Cerebral Edema
- Hypokalemia
- Hypophosphatemia
- Acute Kidney Injury
- Acute Respiratory Distress Syndrome
- Hyperchloremic Acidosis

1.Hypoglycemia

- Glucose value <3.9 mmol/L (70 mg/dl) is used as the clinical alert or threshold value for initiating treatment for hypoglycemia
- Determine underlying cause.
- Monitor blood glucose every 15 minutes until blood glucose is >100 mg/dl.

An international consensus panel

- ✓ **Clinical hypoglycemia alert** – Blood glucose <70 mg/dL (this is commonly used as a threshold for recognizing and initiating treatment for hypoglycemia.
- ✓ **Clinically important or serious hypoglycemia** – Blood glucose <54 mg/dL defective glucose counterregulation, impaired hypoglycemia awareness, and, sometimes, cognitive dysfunction.
- ✓ **Severe hypoglycemia** – An event associated with severe cognitive impairment (including coma and seizures), requiring assistance of another person to correct, including administration of carbohydrates or glucagon or intravenous (IV) dextrose.

Clinical classification of severity

- **Mild** : tremor, palpitations, diaphoresis, headaches, and behavior changes.
older children can treat themselves with oral intake.
- **Moderate** : neurologic symptoms consist of sufficient neurologic impairment
a second person has to administer oral therapy.
- **Severe** : unresponsiveness, seizures, or coma, requires SC administration of glucagon or IV dextrose

cont...

Asymptomatic hypoglycemia:

- Treated orally with a concentrated and rapidly absorbed simple carbohydrate food source
- (10 to 15 g glucose).
- Wait 10–15 minutes
- If no response, repeat above.
- As symptoms improve, the next meal or oral complex carbohydrate should be ingested

Cont....

Symptomatic hypoglycemia:

- Dextrose IV, 2–5 mL/kg of 10% glucose (200 to 500mg/kg). In glucose 10% solution
- IV fluid with concentration of > 10% glucose (glucose concentration = 12%) can be given if no response.
- Monitor blood glucose every 15 minutes until stable, then repeat 1–2 hourly.

Prevention

- Eating meals and snacks on a regular schedule after medications
- Checking blood glucose levels as directed during exercise
- Sticking to a suggested exercise plan
- Carrying juice, candy, or glucose tablets at all times, in the event that one develops symptoms of hypoglycemia

2.Cerebral edema

This complication has a high mortality rate.

Patients who experience cerebral edema with DKA, as many as one-fourth die and another one-fourth have permanent neurologic sequelae.

Poor outcome has been associated with;

- Greater neurologic depression at the time of diagnosis of their cerebral edema,
- Elevated **initial** serum **BUN** concentration, and
- Intubation with hyperventilation to a PCO₂ less than 22 mm Hg (demanding caution in hyperventilating affected patients).

Mechanism of cerebral edema

- The cause of DKA-related cerebral injury is a topic of ongoing investigation.
- Rapid fluid administration resulting in changes in serum osmolality was initially thought to be the cause, however, more recent evidence suggests that cerebral hypoperfusion and the hyperinflammatory state caused by DKA play central roles

Risk factor

Pre treatment

- ✓ *Younger age*
- ✓ *Sick appearance*
- ✓ *New-onset*
- ✓ *Longer duration of symptoms*
- ✓ *More severe evidence of dehydration*
- ✓ *Greater acidosis (lower pCO₂, pH)*

while on treatment

- ✓ *Too-rapid fall glucose*
- ✓ *Failure of sodium to rise*
- ✓ *Too-rapid fall osmolality*
- ✓ *Use of bicarbonate*
- ✓ *Early insulin treatment*
- ✓ *Use of fluids: ≥ 50 mL/kg in the first 4 hours*

Signs and symptoms

- sudden deterioration in mental status
- severe headache
- sudden drop in heart rate not attributed to rehydration
- hypertension, cranial nerve dysfunction, and incontinence.
- Cushing's triad(late sign)
- Decreased O₂ saturation

Diagnosis

- **Diagnostic criteria**
 - Abnormal motor or verbal response to pain
 - Decorticate or decerebrate posture
 - Cranial nerve palsy (especially III, IV, and VI)
 - Abnormal neurogenic respiratory pattern (e.g., grunting, tachypnea, apneusis, Cheyne-Stokes respiration)
- **Minor criteria**
 - Vomiting
 - Headache
 - Lethargy or not easily arousable
 - Diastolic blood pressure > 90 mmHg
 - Age <5 years
- **Major criteria**
 - Altered mentation, confusion, fluctuating level of consciousness
 - Sustained heart rate deceleration not attributable to improved intravascular volume or sleep state
 - Age-inappropriate incontinence
- **Treat**
 - One Diagnostic criteria
 - two major criteria,
 - one major and two minor criteria

Management

- Reduction in the rate of fluid administration
- Elevation of the head of the bed
- Administration of intravenous mannitol in a dosage of 0.25 to 1.0 g/kg over 10-15 minutes, repeated as necessary after 30min.
- Hypertonic saline (3%), suggested dose 2.5 to 5 mL/kg over 10 to 15 minutes
 - may be used as an alternative to mannitol, **or**
 - in addition to mannitol if there has been no response to mannitol within 15 to 30 minutes.
- Hypertonic saline (3%) 2.5 mL/kg is equimolar to mannitol 0.5 g/kg

Effects of manitol

- lowering hematocrit and blood viscosity, which improves cerebral blood flow (CBF) and oxygenation
- Improvement in red cell deformability
- vasoconstriction in areas of the brain with intact autoregulation or improved autoregulation from the effects of the mannitol on CBF and oxygenation; and
- Direct osmotic effects with reduction in extracellular free water.
- Intubation - for those who have progressed to respiratory compromise

3. Akute Kidney Injury

- Can occur upto 43 to 64 percent
- Many of these children have stage 2 or 3 AKI, suggesting intrinsic tubular injury beyond prerenal dysfunction
- Renal failure can also occur but is rare
- AKI generally resolves after recovery from DKA

4.Acute Respiratory Distress Syndrome

- Non-cardiogenic pulmonary edema results from reduced colloidal oncotic pressure in pulmonary capillaries due to crystalloid replacements in DKA.
- Presence of hypoxemia with elevated pulmonary alveolar-arterial gradient, warrant suspicion
- It can be avoided by judicious fluid replacement.

5. Hyperchloremic Acidosis

- The use of large amounts of chloride-rich fluids may be associated with the rapid development of hyperchloremia
- It is defined as $[\text{Cl}^-:\text{Na}^+] > 0.79$ and hyperchloremic metabolic acidosis.
- Hyperchloremic acidosis resolves spontaneously

Phase III :continuing or maintenance phase.

Includes the insulin regimen after the patient has maintained the target RBS for the transition phase in the past 24 hours.

- i) In **Newly** diagnosed DM: $\frac{2}{3}$ of the calculated total subcutaneous insulin in the past 24 hours will be the representative daily dose in the maintenance phase.
- ii) For **known** DM patients : the calculated total subcutaneous insulin in the past 24 hours will be the representative daily dose in the maintenance phase

Cont...

- Out of the total daily insulin requirement, 2/3 will be given in the morning & the remaining 1/3 in the evening.
- And 2/3 will be intermediate acting & 1/3 fast acting insulin.
- The ideal target RBS during this regimen is **80mg/dl for fasting & 140mg/dl** after meal; but in practice a range of 60 – 220mg/dl is acceptable based on the age of the patient.
- RBS measurements that are constantly out of this range without identifiable cause, are indications for adjusting insulin doses;i.e.by increasing or decreasing **the next insulin dose by 10 – 15%.**

Reference

- ISPAD 2022
- Uptodate 2026
- Nelson 21st edition